

Lancaster University College
at Beijing Jiaotong University

Digital Signal Processing

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Implementation of digital filter

- The basic structures of IIR digital filters
- The basic structures of FIR digital filters
- Finite word length effects

The basic structures of IIR digital filters

- Direct form structure
- Cascaded form structure
- Parallel form structure

Direct form structure of IIR digital filters

- Based on the difference equation describing the discrete LTI (Linear Time-Invariant) system:

$$y[k]+\sum_{j=1}^{N} a_j y[k-j]=\sum_{i=0}^{M} b_i x[k-i]$$

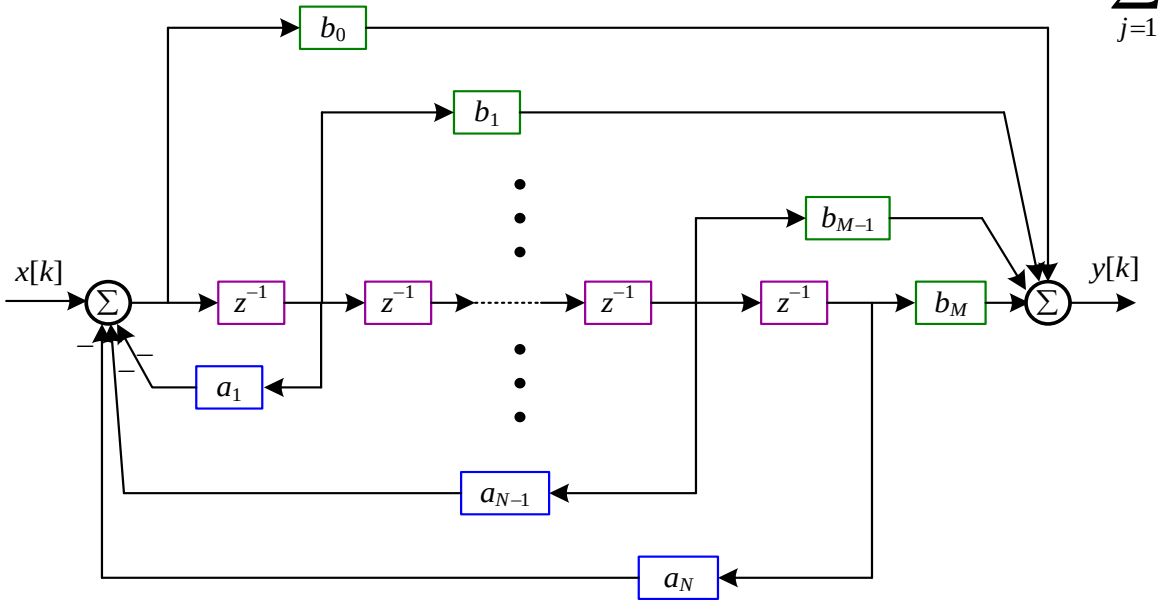
- The system function H(z) of the discrete LTI system can be obtained as:

$$H(z)=\frac{b_0+b_1 z^{-1}+\cdots+b_{M-1}z^{-(M-1)}+b_M z^{-M}}{1+a_1 z^{-1}+\cdots+a_{N-1}z^{-(N-1)}+a_N z^{-N}}=\frac{\sum_{i=0}^{M} b_i z^{-i}}{1+\sum_{j=1}^{N} a_j z^{-j}}$$

The direct form structure of IIR digital filter

Let $M = N$

$$H(z) = \frac{b_0 + b_1 z^{-1} + \cdots + b_{M-1} z^{-(M-1)} + b_M z^{-M}}{1 + a_1 z^{-1} + \cdots + a_{N-1} z^{-(N-1)} + a_N z^{-N}} = \frac{\sum_{i=0}^{M} b_i z^{-i}}{1 + \sum_{j=1}^{N} a_j z^{-j}}$$



The direct form structure of IIR digital filter

$$H(z)=\frac{b_0+b_1z^{-1}+\cdots+b_{M-1}z^{-(M-1)}+b_Mz^{-M}}{1+a_1z^{-1}+\cdots+a_{N-1}z^{-(N-1)}+a_Nz^{-N}}$$

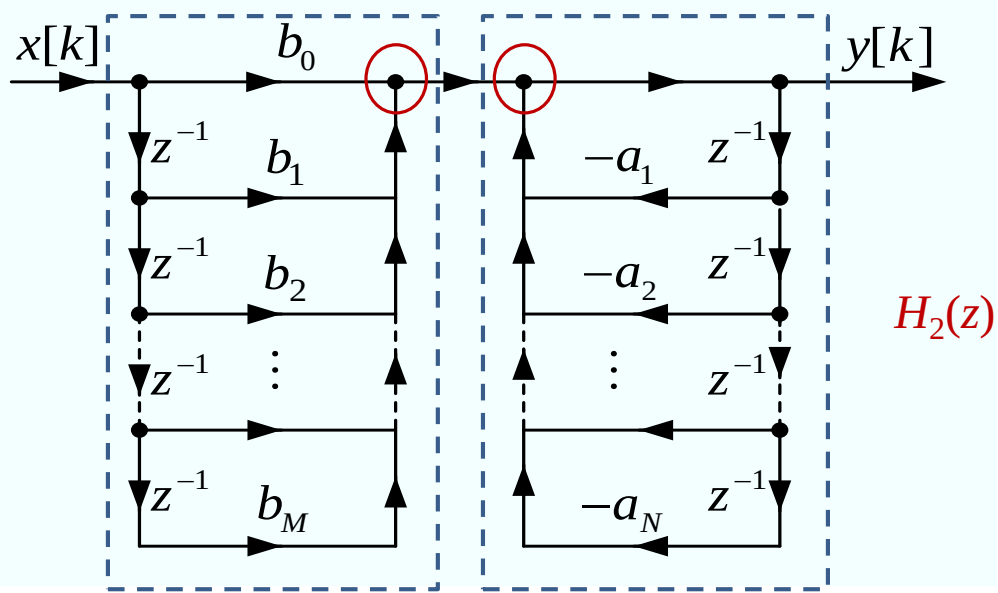
$$=\sum_{i=0}^{M}b_iz^{-i}\cdot\frac{1}{1+\sum_{j=1}^{N}a_jz^{-j}}$$

$H_1(z)$ $H_2(z)$

The direct form structure of IIR digital filter

※ Direct Form I Structure

$Let\ M = N$

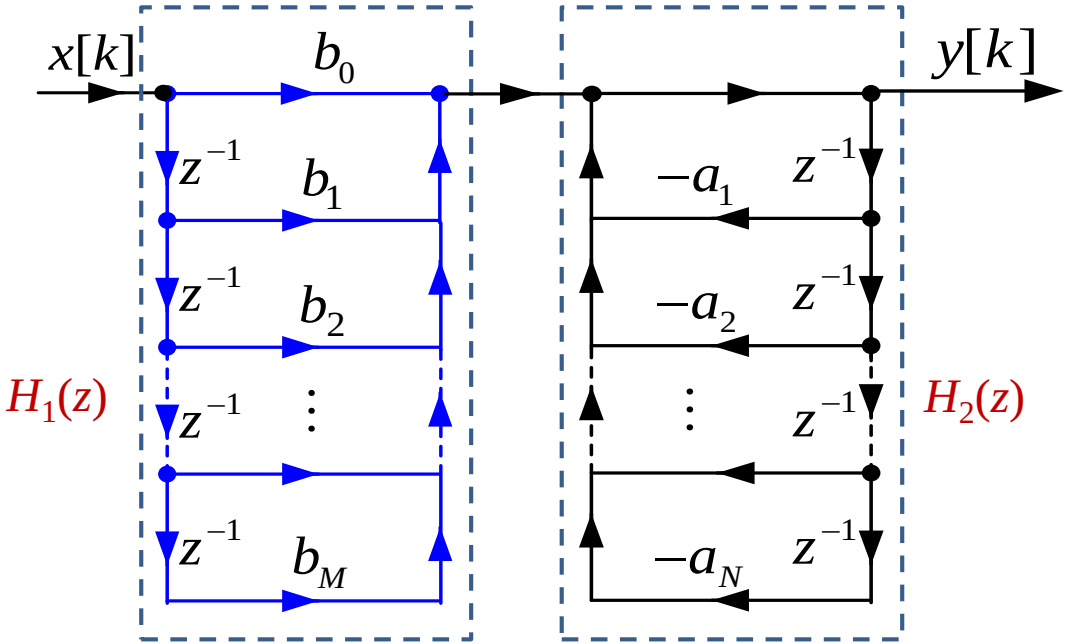


$$H_1(z) = \sum_{i=0}^{M} b_i z^{-i}$$

$$H_2(z) = \frac{1}{1 + \sum_{j=1}^{N} a_j z^{-j}}$$

The direct form structure of IIR digital filter

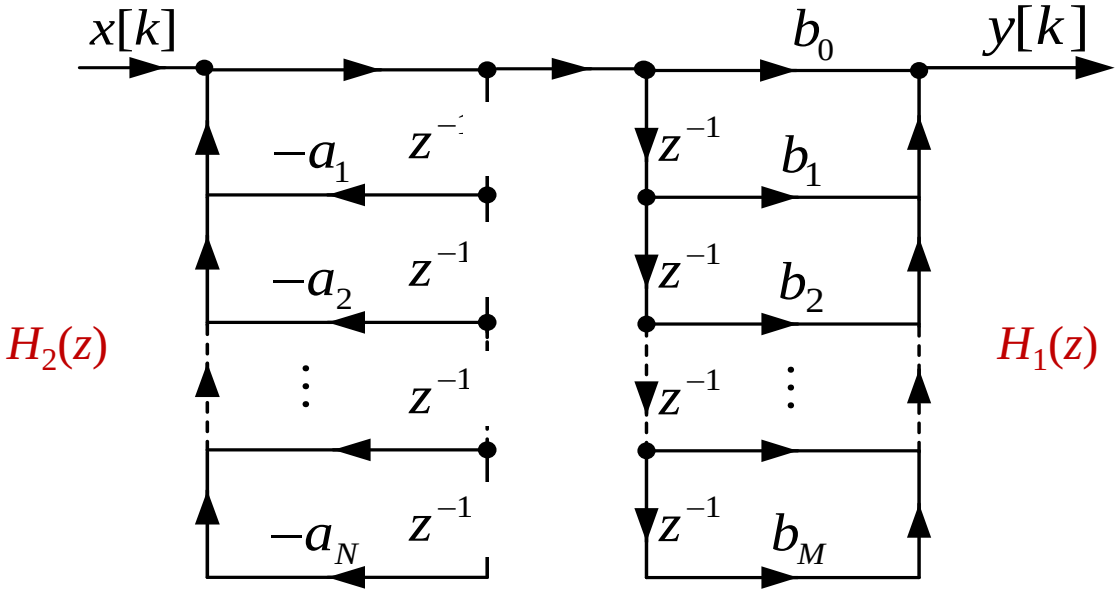
※ Direct Form II Structure



swap the cascade order of $H_1(z)$ and $H_2(z)$

The direct form structure of IIR digital filter

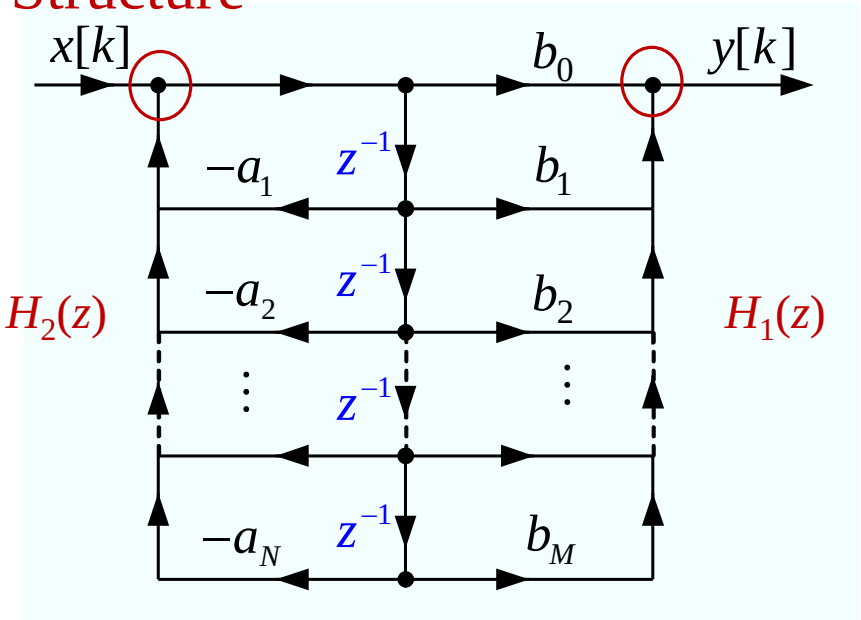
※ Direct Form II Structure



$H_2(z)$ and $H_1(z)$ share delay elements.

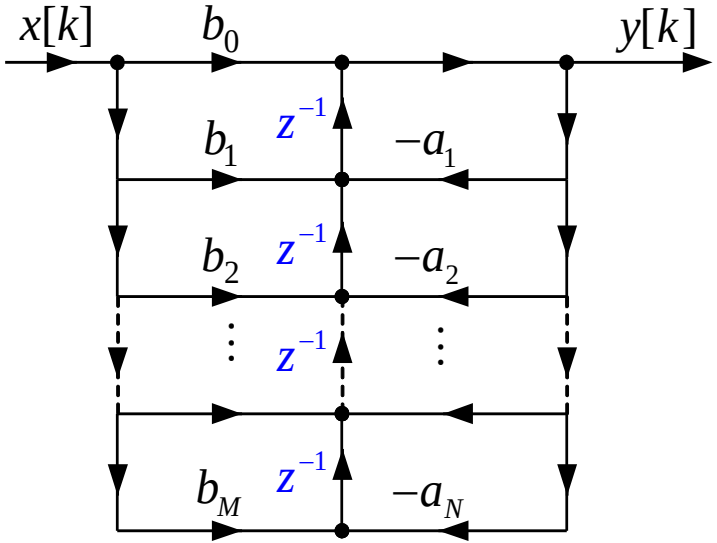
The direct form structure of an IIR digital filter

※ Direct Form II Structure

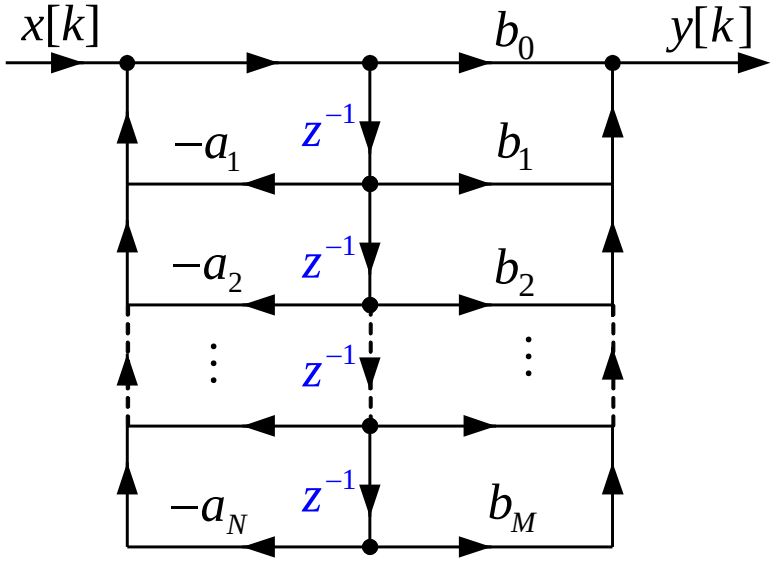


The direct form structure of IIR digital filter

※Transposed Direct Form II Structure



Transposed Direct Form II Structure



Direct Form II Structure

The direct form structure of IIR digital filter

※ The characteristics of the direct form structure of an IIR digital filter are:

Advantages: Simple and intuitive

Disadvantages:

1. Changing a single $\{a_k\}$ will affect all poles.
2. Changing a single $\{b_k\}$ will affect all zeros.
3. It is very sensitive to finite word length effects, which can easily lead to instability.

For IIR filters of third order and above, the direct form structure is generally not used. Instead, other structures such as cascade or parallel forms are typically employed.

The cascade structure of IIR digital filter

If the numerator and denominator of the digital filter system function H(z) are factored into a product of first-order or second-order real coefficient factors, and the first-order factors are represented as second-order factors:

Second-order section

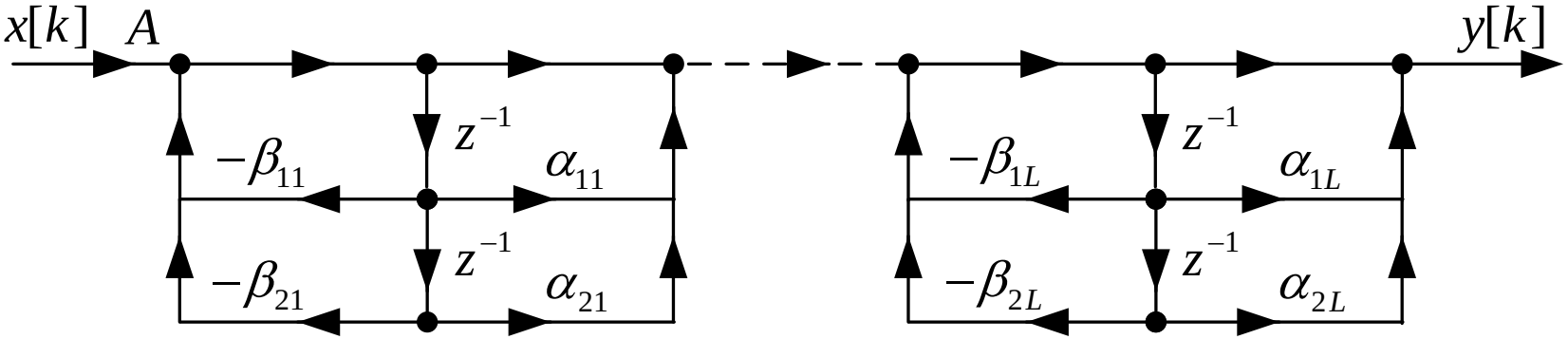
$$H(z)=\prod_{i=1}^{L}\frac{\gamma_{0i}+\gamma_{1i}z^{-1}+\gamma_{2i}z^{-2}}{1+\beta_{1i}z^{-1}+\beta_{2i}z^{-2}}=A\prod_{i=1}^{L}\frac{1+\alpha_{1i}z^{-1}+\alpha_{2i}z^{-2}}{1+\beta_{1i}z^{-1}+\beta_{2i}z^{-2}}=A\prod_{i=1}^{L}H_i(z)$$

Draw the direct form structure of each second-order section, and then cascade them to form the cascade structure.

The cascade structure of IIR digital filter

※ Cascade structure based on Direct Form II

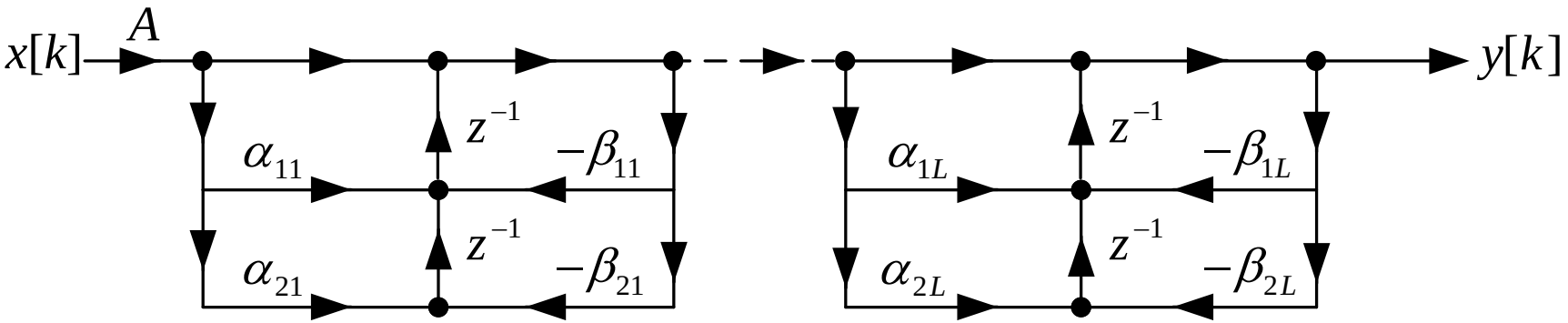
$$H(z) = A\prod_{i=1}^{L}\frac{1+\alpha_{1i}z^{-1}+\alpha_{2i}z^{-2}}{1+\beta_{1i}z^{-1}+\beta_{2i}z^{-2}}$$



The cascade structure of IIR digital filter

※ Cascade structure based on the transposed Direct Form II

$$H(z) = A\prod_{i=1}^{L}\frac{1+\alpha_{1i}z^{-1}+\alpha_{2i}z^{-2}}{1+\beta_{1i}z^{-1}+\beta_{2i}z^{-2}}$$



The cascade structure of IIR digital filter

The cascade structure of an IIR digital filter: $H(z) = A\prod_{i=1}^{L}\frac{1+\alpha_{1i}z^{-1}+\alpha_{2i}z^{-2}}{1+\beta_{1i}z^{-1}+\beta_{2i}z^{-2}}$

- The coefficient changes in each individual section only affect the zeros and poles of that specific subsystem.
- The zeros and poles of each subsystem are the zeros and poles of the entire system, making it easier to adjust the system characteristics
- The sensitivity to coefficient changes is low, and it is less affected by finite word length effects compared to the direct form.
- The cascade order of each second-order section can be adjusted, and the pairing of zeros and poles can also be changed. Therefore, the cascade structure for the same system function is not unique.

The parallel structure of IIR digital filter

If the filter system function $H(z)$ is expanded into a sum of partial fractions, and each first-order subsystem is represented by a second-order section:

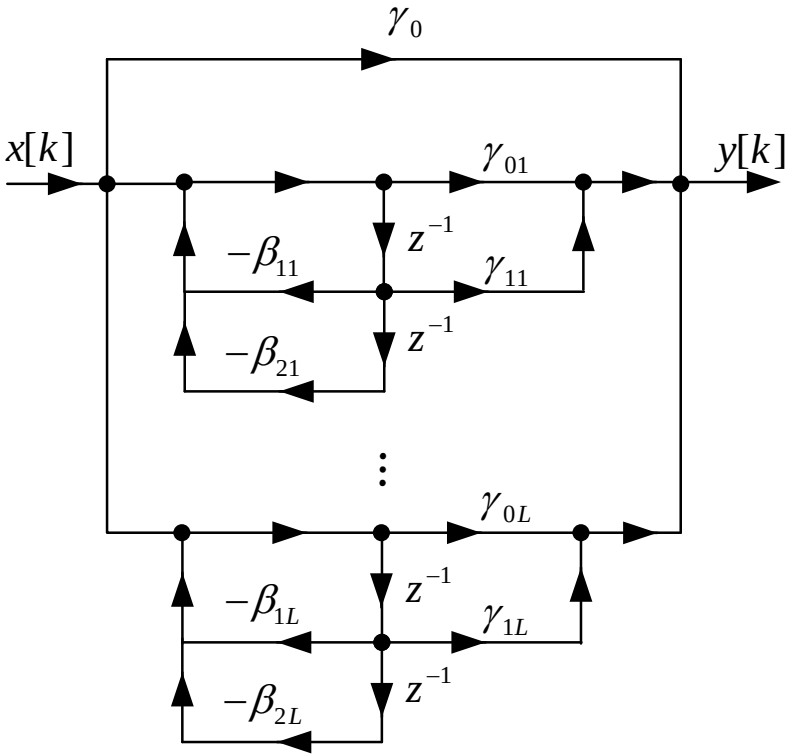
$$H(z) = \gamma_0 + \sum_{k=1}^{L} \frac{\gamma_{0k} + \gamma_{1k} z^{-1}}{1 + \beta_{1k} z^{-1} + \beta_{2k} z^{-2}}$$

Draw the direct form structure of each second-order section, and then parallel them to form the parallel structure.

The parallel structure of IIR digital filter

※ Parallel structure based on Direct Form II

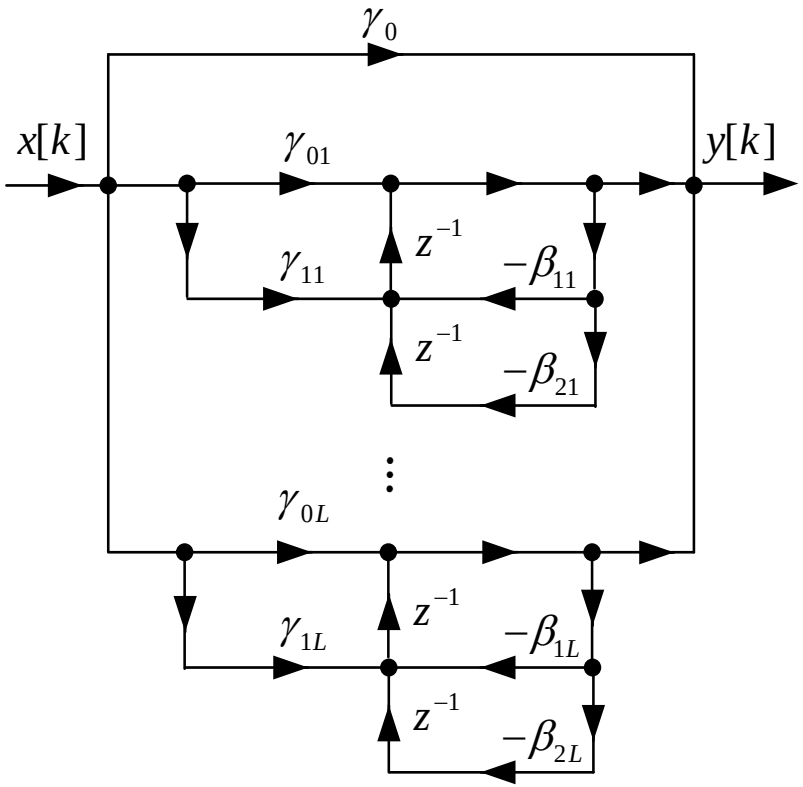
$$H(z)=\gamma_0+\sum_{k=1}^{L}\frac{\gamma_{0k}+\gamma_{1k}z^{-1}}{1+\beta_{1k}z^{-1}+\beta_{2k}z^{-2}}$$



The parallel structure of IIR digital filter

※ Parallel structure based on the transposed Direct Form II

$$H(z)=\gamma_0+\sum_{k=1}^{L}\frac{\gamma_{0k}+\gamma_{1k}z^{-1}}{1+\beta_{1k}z^{-1}+\beta_{2k}z^{-2}}$$



The parallel structure of IIR digital filter

※ Characteristics of the parallel structure of an IIR digital filter:

$$H(z)=\gamma_0+\sum_{k=1}^{L}\frac{\gamma_{0k}+\gamma_{1k}z^{-1}}{1+\beta_{1k}z^{-1}+\beta_{2k}z^{-2}}$$

- High computation speed and short processing delay
- Errors in each section do not affect each other
- Changes in the coefficients of each section only affect the zeros and poles of that specific subsystem
- The zeros cannot be directly adjusted, and the zeros of the subsystems are not the zeros of the entire system

Examples of IIR digital filter structures

[Example] Given the system function of a third-order digital filter as:

$$H(z)=\frac{3+\frac{5}{3}z^{-1}+\frac{2}{3}z^{-2}}{(1-\frac{1}{3}z^{-1})(1+\frac{1}{2}z^{-1}+\frac{1}{2}z^{-2})}$$

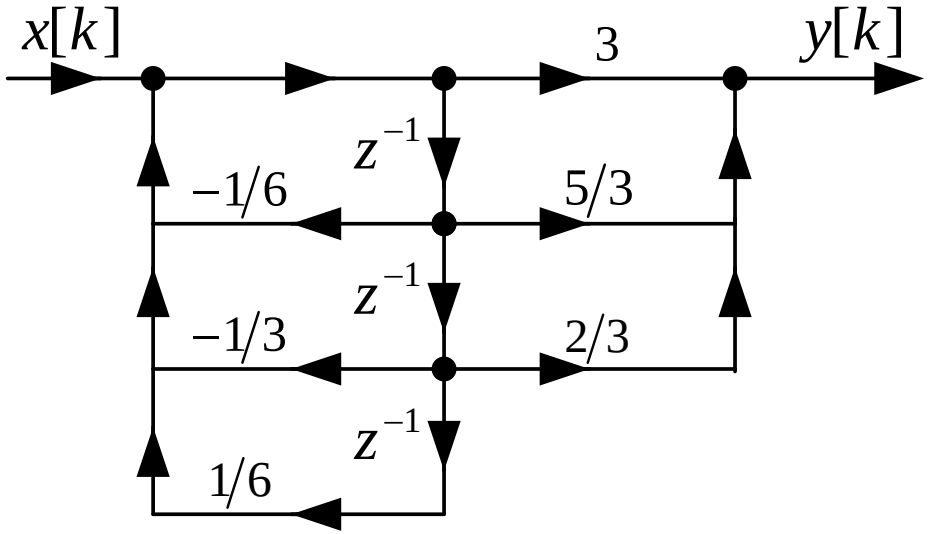
Try to draw its direct form, cascade form, and parallel form structures.

Examples of IIR digital filter structures

Solution: Direct Form II Structure

Express the system function $H(z)$ as:

$$H(z)=\frac{3+\frac{5}{3}z^{-1}+\frac{2}{3}z^{-2}}{1+\frac{1}{6}z^{-1}+\frac{1}{3}z^{-2}-\frac{1}{6}z^{-3}}$$

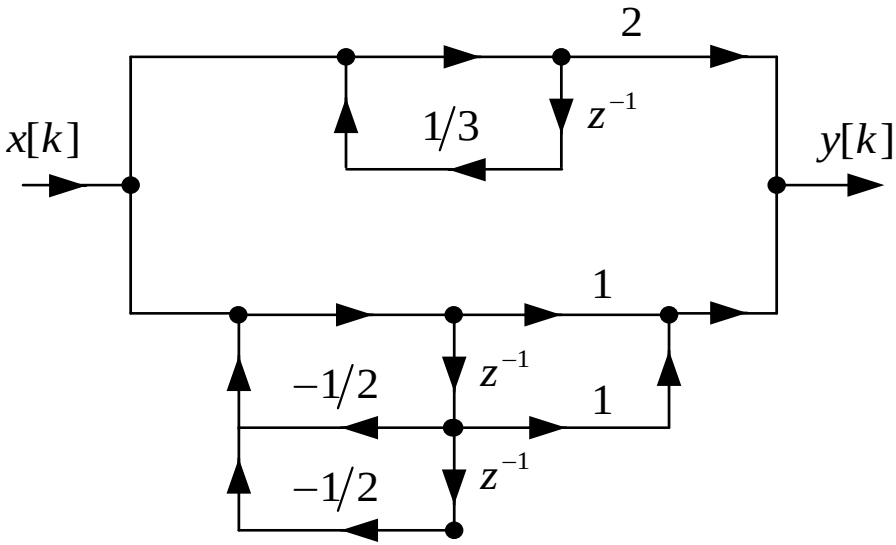


Examples of IIR digital filter structures

Solution: Parallel Form

Express the system function $H(z)$ as:

$$H(z)=\frac{2}{1-\frac{1}{3}z^{-1}}+\frac{1+z^{-1}}{1+\frac{1}{2}z^{-1}+\frac{1}{2}z^{-2}}$$

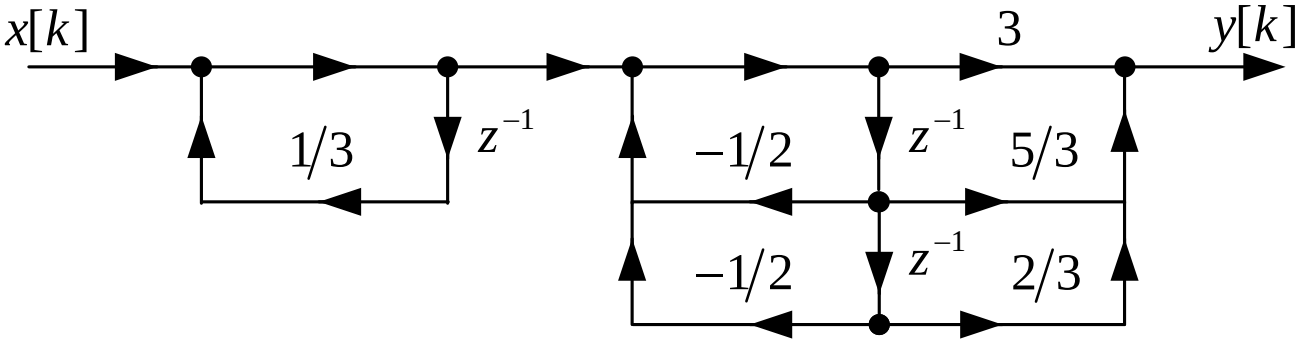


Examples of IIR digital filter structures

Solution: Cascade Form

Express the system function $H(z)$ as:

$$H(z)=\frac{1}{1-\frac{1}{3}z^{-1}}\cdot\frac{3+\frac{5}{3}z^{-1}+\frac{2}{3}z^{-2}}{1+\frac{1}{2}z^{-1}+\frac{1}{2}z^{-2}}$$



Use MATLAB to implement a digital filter

[Example] A 4th-order Butterworth (BW) IIR digital band-stop filter is as follows:

$$H(z)=\frac{0.9522+3.7326z^{-1}+5.5624z^{-2}+3.7326z^{-3}+0.9522z^{-4}}{1+3.8241z^{-1}+5.5601z^{-2}+3.6412z^{-3}+0.9067z^{-4}}$$

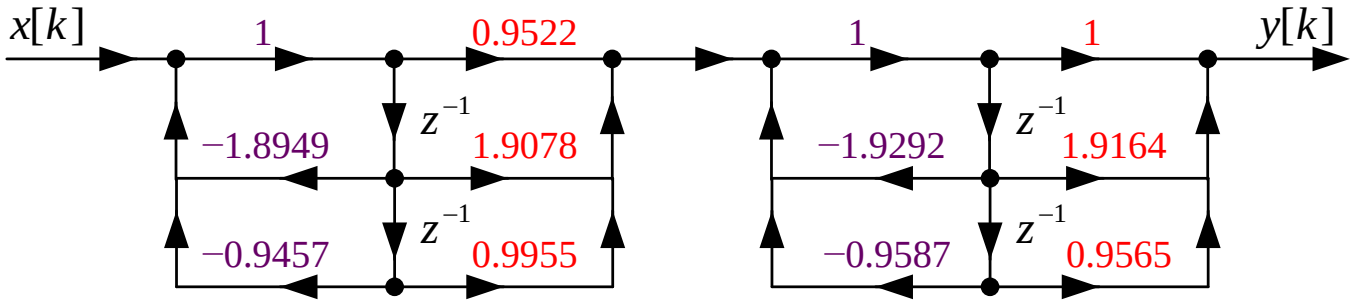
Try to find the coefficients of the cascade structure for this filter.

```
num = [0.9522   3.7326   5.5624 3.7326   0.9522];
den = [1   3.8241  5.5601   3.6412 0.9067];
sos = tf2sos(num,den);     %  Obtain the coefficients for the second-order sections
[row col] = size(sos);
for i = 1:row
    for j = 1:col
        fprintf('%2.4f\t', sos(i, j));
    end
end
```

Use MATLAB to implement a digital filter

Cascade structure IIR digital filter coefficients:

Subsystem 1	0.9522	1.9078	0.9955	1.0000	1.8949	0.9457
Subsystem 2	1.0000	1.9164	0.9565	1.0000	1.9292	0.9587



Use MATLAB to implement a digital filter

[Example] A 4th-order Butterworth (BW) IIR digital band-stop filter is as follows:

$$H(z)=\frac{0.9522+3.7326z^{-1}+5.5624z^{-2}+3.7326z^{-3}+0.9522z^{-4}}{1+3.8241z^{-1}+5.5601z^{-2}+3.6412z^{-3}+0.9067z^{-4}}$$

Try to find the coefficients of the parallel structure for this filter

```
num = [0.9522   3.7326   5.5624   3.7326   0.9522];
den =  [1       3.8241   5.5601   3.6412   0.9067];
[A p K] = residuez(num, den);
```

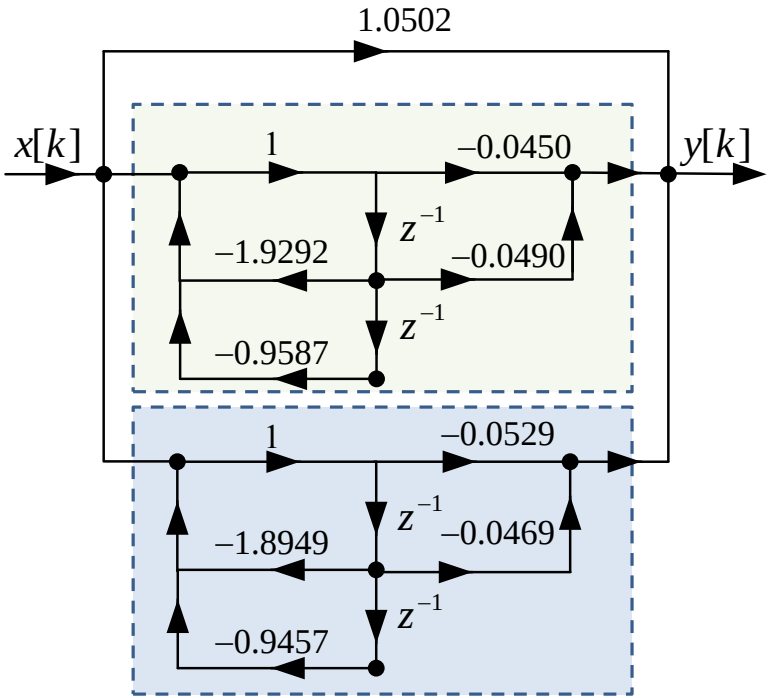
$$H(z)=K+\sum_{i=1}^{L}\frac{A_i}{z-p_i}$$

Parallel structure IIR digital filter coefficients:

Coefficient A:	pole p:
– 0.0225 + j0.0166	– 0.9646 + j0.1680
– 0.0225 – j0.0166	– 0.9646 – j0.1680
– 0.0265 – j0.0075	– 0.9474 + j0.2193
– 0.0265 + j0.0075	– 0.9474 – j0.2193

Constant term K: 1.0502

Use MATLAB to implement a digital filter



Coefficient A:	pole p:
–0.0225 + j0.0166	– 0.9646 + j0.1680
–0.0225 – j0.0166	– 0.9646 – j0.1680
–0.0265 – j0.0075	– 0.9474 + j0.2193
–0.0265 + j0.0075	– 0.9474 – j0.2193

```
[a1,b1] = residuez(A(1:2),p(1:2),[]);
[a2,b2] = residuez(A(3:4),p(3:4),[]);
```

```
b1 = –0.0450   –0.0490
a1 = 1.0000    1.9292    0.9587
```

```
b2 = –0.0529   –0.0469
a2 = 1.0000    1.8949    0.9457
```

Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!